



Research paper

Cancer–drug response prediction via feature aggregation and association graph learning

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ABSTRACT

Predicting cancer–drug response (CDR) not only guides the development of anticancer drugs but also provides personalized treatment plans for cancer patients, driving the advancement of precision medicine. Currently, numerous computational methods are utilized for CDR prediction and achieve great achievements. However, there are still some issues need to be addressed. From the aspect of data using, many studies primarily use single nucleotide variations (SNV) as features of cell lines, yet their acquisition from clinical data is often incomplete and inefficient, making it challenging to extend SNV-based CDR models from in vitro cell lines to clinical data. From the aspect of modeling, most works focus excessively on learning the intrinsic characteristics of drugs and cell lines, while neglecting to fully capture their interaction information and network-level knowledge. In this paper, we introduce a feature aggregation and association graph learning network (FA²GL) for CDR, which aggregates intrinsic and interaction feature representations of drugs and cell lines, and obtains their network-level knowledge from the association graph. In this network, FA²GL initially employs the self-attention block (SAB) to capture drug intrinsic features, with a focus on the relationships between substructures within the same drug, and utilizes the multilayer perceptron (MLP) to obtain cell line intrinsic features. Subsequently, it constructs the cross-attention block (CAB) to extract interaction information between drugs and cell lines, followed by the utilization of a fusion coefficient for aggregating these features. Additionally, FA²GL also constructs an association graph by introducing pipe nodes and similarity networks, bridging the feature gap between drugs and cell lines and enhancing predictions of similar substances, respectively. Then, a graph neural network (GNN) is applied to this graph to capture their network-level information. By integrating fine-grained and network-level features, FA²GL demonstrates competitiveness compared to state-of-the-art methods on Genomics of Drug Sensitivity in Cancer (GDSC) dataset which contains various cancers such as acute lymphoblastic leukemia, diffuse large B-Cell lymphoma, and multiple myeloma, while avoiding the use of large-scale SNV data. The codes and data for FA²GL are available at <https://github.com/Dtdtxuky/FA-2GL/tree/main>.

1. Introduction

Cancer is a leading cause of death globally (Angelini et al., 2023), and anticancer drugs are pivotal in reducing mortality rates among cancer patients (Zhang et al., 2020). However, due to the heterogeneity within and between tumors, patients may exhibit varying responses to anticancer drugs (Kohane, 2015). Studying CDR not only accelerates the screening and development of anticancer drugs but also promotes the implementation of personalized medicine by assessing the effectiveness of specific drugs based on patients' genetic profiles.

The CDR can be studied using various in vitro cancer cell line models (Grever et al., 1992). Researchers quantify drug sensitivity to cancer cells through the half-maximal inhibitory concentration (IC₅₀). The smaller the IC₅₀ value, the greater the inhibitory effect of the drug on the cancer cells, and vice versa (Sebaugh, 2011). Nowadays, rapid development in high-throughput technologies has significantly advanced CDR research (Reuter et al., 2015; Soon et al., 2013). For example, the widespread use of sequencing technologies allows for comprehensive analysis of cancer cells across various omics profiles,

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including genomics (e.g. copy number variations (CNV)), transcriptomics (e.g. gene expression), and epigenomics (e.g. DNA methylation), enhancing our understanding of the genetic and epigenetic features of cancer cells (Balmain et al., 2003; Fan et al., 2020; Jones and Baylin, 2007). Moreover, researchers also employ high-throughput screening techniques to evaluate the effects of chemical compounds from libraries on multiple cancer cell lines cultured under diverse conditions and determine their corresponding IC50 values. Expanding the scale of screening experiments would greatly benefit precision medicine. However, this requires larger-scale experimental equipment, which results in higher costs. Additionally, due to resource and technological constraints, CDR matrices from large-scale high-throughput screening are often incomplete. Therefore, developing an accurate and cost-effective tool to predict all IC50 values is crucial for anticancer drug screening (Menden et al., 2013).

Currently, multiple pharmacogenomic datasets have been established, including the Genomics of Drug Sensitivity in Cancer (GDSC) (Yang et al., 2012), and Cancer Cell Line Encyclopedia (CCLE) (Barretina et al., 2012). Based on these datasets, researchers have started utilizing machine learning techniques to predict CDR, including support vector machine (Dong et al., 2015), random forest (Riddick et al., 2011; Iorio et al., 2016), and logistic ridge regression (Geeleher et al., 2014). While these methods have shown promise in predicting CDR, they still face limitations when handling cancer cell line data: (1) These methods require prior feature selection of those that are closely linked to specific biological processes. (2) High-dimensional genomic data may lead these models to overfit. (3) The integration of multi-omics information remains insufficient.

Compared to machine learning, deep learning techniques not only automatically extract efficient features but also demonstrate remarkable performance in feature fusion and handling high-dimensional data, such as images, text, and spectra (Tang et al., 2023b; Wang et al., 2023; Vaswani et al., 2017; Tang et al., 2023a). Currently, deep learning methods for CDR prediction can be categorized into two categories: network-based and feature learning-based methods.

The former approach, similar to recommendation systems, relies on similarity and association information to make predictions. For instance, (Zhang et al., 2015) proposed a dual-association network using the similarity of drugs and cancer cell lines to predict CDR. In addition, DREMO (Yu et al., 2021) utilized a multi-view, multi-layer similarity network to derive low-dimensional vector representations of drugs and cell lines.

In contrast, the latter approach delves into the internal characteristics of drugs and cell lines by analyzing features of atoms, chemical bonds, genes, and transcription products. For example, CDRscan (Chang et al., 2018) utilized convolutional neural networks (CNNs) to encode molecular fingerprints of drugs and somatic mutations of cell lines. Similarly, tCNNs (Liu et al., 2019) also leveraged CNNs to process Simplified Molecular Input Line Entry System (SMILES) sequences of drugs and genomic mutation data of cell lines. To explore deeper into the internal topological features of drugs, DeepCDR (Liu et al., 2020) employed GNNs (Scarselli et al., 2008) to manipulate drug molecular graphs, which treats atoms as nodes and chemical bonds as edges. Building upon this approach, DualGCN (Ma et al., 2022) constructed gene relationship graphs within cell lines, incorporating gene expression and CNV as node features, and interactions of protein–protein guided by genes as edges.

Based on the powerful representational capabilities of deep learning technology, above methods have shown outstanding performance in predicting CDR, yet they also have certain limitations:

- SNV, reflecting the variations of individual nucleotides within oncogenes, is utilized in certain models as cell line features. However, due to the typically lower sequencing coverage and depth of single-cell data, SNV's detection often suffers from incompleteness (Gawad et al., 2016). Additionally, the high-frequency

genomic aberrations and aneuploidy commonly observed in cancer cells further reduce the efficiency of SNV's detection (Adey et al., 2013).

- Above models overly emphasize the intrinsic information of drugs and cell lines while overlooking their interaction features, which limits the comprehensive understanding of CDR.
- Most feature learning-based models (Liu et al., 2019; Ma et al., 2022; Jiang et al., 2022) fail to consider the inherent similarities within drugs and cell lines, which could not heuristically enhance the predictive results for similar substances.

To address the aforementioned issues, We propose a novel feature aggregation and association graph learning network, namely FA²GL. Its specific contributions are as follows:

- FA²GL captures intrinsic features and interaction information of drugs and cell lines through SAB, MLP, and CAB respectively, and aggregates them using a fusion coefficient as fine-grained features, achieving a comprehensive representation of drugs and cell lines at the individual level.
- We construct an association network graph by introducing virtual pipeline nodes to connect drugs and cell lines and utilizing similarity as edges between homogeneous nodes. This approach not only bridges the feature gap between drugs and cell lines but also enhances the representation of similar substances.
- FA²GL combines two approaches for predicting CDR: feature learning-based and network-based methods, taking fine-grained features and network-level characteristics into consideration. This combination of different granularity features contributes to FA²GL's excellent performance in CDR prediction.

2. Related works

In this section, we provide an overview of related works and techniques in drug–cancer response prediction.

2.1. Drug–cancer response prediction

Drug–cancer response prediction is a critical area in precision medicine, aiming to predict the sensitivity of cancer cells or tumors to specific drugs based on their molecular characteristics. By accurately forecasting how cancer responds to drugs, clinicians can not only devise personalized treatment strategies but also provide robust support for new drug development and drug repositioning. With the continuous accumulation of large-scale drug response datasets (e.g., GDSC, CCLE), data-driven computational methods are increasingly applied in this field (Barretina et al., 2012; Garnett et al., 2012), and deep learning has also emerged as a research hotspot (Zaki et al., 2021).

2.2. Graph Neural Networks

Graph Neural Networks (GNNs) have been widely applied in various fields, including bioinformatics, social networks, chemistry, and recommendation systems, due to their ability to process unstructured data and capture hidden relational structures (Ying et al., 2018; Wu et al., 2018; Kojima et al., 2020). In bioinformatics, GNNs have been utilized to handle complex network structures and heterogeneous data, such as protein–protein interaction networks, molecular graphs, and gene regulatory networks, achieving promising results (Zaki et al., 2021; Gao et al., 2021). HMM-GDAN (Liu et al., 2023) is a hybrid multi-view and multi-scale graph duplex-attention network for cancer–drug response prediction. This model integrates multi-omics data and protein–protein interaction information through multi-view self-attention and view-level attention mechanisms to capture the complex relationships between drugs and cell lines. DualGCN (Ma et al., 2022) employs a dual-channel graph convolutional network to simultaneously capture drug and cell line features, enhancing the accuracy of drug response

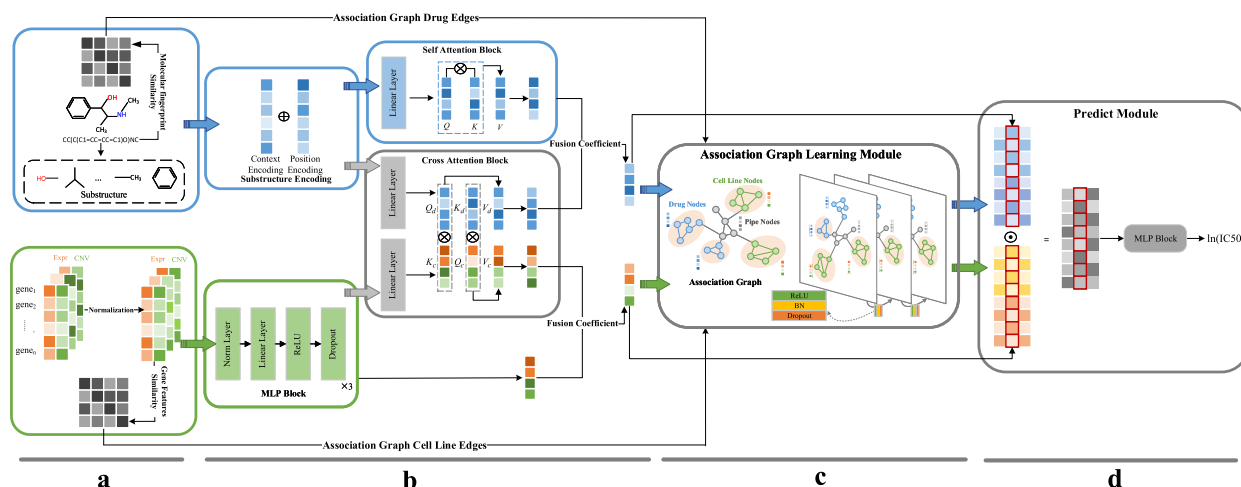


Fig. 1. The workflow of FA²GL. (a) Data preprocessing: Drugs are decomposed into substructures based on their SMILES representations, and then a similarity matrix is generated based on drugs' Morgan fingerprints. For cell lines, gene expression and CNV data are normalized, and a cosine similarity matrix between cell lines is computed using gene vectors. (b) Feature representation and aggregation: After that, drug features are derived by encoding substructures' content and position, and gene vectors are inputted into an MLP to generate embeddings of cancer cell lines. These embeddings undergo two key processes: (1) They are fed into the CAB to obtain interaction features, and (2) drugs' embeddings are further refined within the SAB. Fusion coefficients are then applied to integrate both intrinsic and interaction features. (c) Association graph learning module: Afterwards, additional pipe nodes are introduced to collectively construct an association graph with multiple drug and cell line nodes. Then, we utilize GNNs to capture their network features on the above graph. (d) Predict module: Finally, the features of drugs and cell lines, which are concatenated results from features (b) and (c), are multiplied pointwise and inputted into an MLP block to predict $\ln(\text{IC}_{50})$ values.

prediction. Additionally, MOFGCN (Peng et al., 2021) integrates multi-omics data, including gene expression, copy number variation, and somatic mutations, using a graph convolutional network to process and aggregate this information for drug response prediction. These methods leverage the advantages of graph neural networks in handling complex biological data, offering new perspectives for personalized cancer treatment.

3. Method

We propose an end-to-end network for predicting CDR, named FA²GL. Its framework is illustrated in Fig. 1. FA²GL takes drugs' SMILES and cell lines' gene expression and CNV as inputs, and predicts the $\ln(\text{IC}_{50})$. It comprises four parts: (1) Data preprocessing: It utilizes the open-source library subword_nmt to decompose drugs' SMILES string into substructure sequences and constructs a similarity matrix based on their Morgan fingerprints. In addition, gene data of cell lines is normalized, and a similarity matrix of cell lines is also constructed. (2) Feature representation and aggregation: the SAB and MLP are applied to extract semantic features of drug substructures and gene features of cell lines from drugs' substructure sequences and normalized gene data, respectively. CAB is employed to obtain interaction representations between them. Then, above features are aggregated by a fusion coefficient as fine-grained features. (3) Association graph learning module: FA²GL introduces pipe nodes and utilizes similarity matrices to construct an association graph between drugs and cell lines. Thus, we can leverage GNNs to capture network-level features on the graph. (4) Predict module: The fine-grained and network-level features are concatenated and inputted into an MLP to produce the final predicted values of $\ln(\text{IC}_{50})$.

3.1. Data preprocessing

The functional groups/chemical substructures determine all pharmacokinetic and pharmacodynamic properties of drugs (Kofuji et al., 2005). Therefore, instead of constructing molecular graphs, we opt to decompose drugs into substructures based on their SMILES. These SMILES strings consist of a series of characters representing atoms, bonds, and rings, conveying structural, connective, and chemical semantic information. In this work, we utilize the open-source library

subword_nmt to decompose drugs into substructures. The core idea is to split SMILES strings into characters and iteratively merge the most frequently occurring 2-g until reaching a specified vocabulary size. For example, 'CC(C(C)NCC(COC1=CC=CC=C1C2CCCC2)O)', the desipramine's SMILES string, is decomposed into ['CC(C(C)', 'NCC', '(C', 'OC1', '=CC', '=CC=C', '1C', '2CCCC', '2)', 'O']. Based on the maximum number of substructures per drug, we standardize the number for all drugs to 50, filling missing slots with zeros as needed. Additionally, we use the Morgan fingerprints of drugs to compute the Tanimoto similarity coefficient for constructing a drug similarity matrix. Regarding cancer cell lines, we normalize the gene data, including gene expression and CNV, using mean-variance normalization. For each gene (or CNV feature) in the dataset, we compute the normalized value as follows:

$$x = \frac{x - \mu}{\sigma} \quad (1)$$

where μ is the mean and σ is the standard deviation of the gene expression values, respectively, across all cell lines for a given gene (or CNV feature). After that, we calculate cosine similarity to derive the cell line similarity matrix.

3.2. Feature representation and aggregation

In this subsection, we will discuss the following three processes: (1) Encoding the content and position of drug substructures from Section 2.1, followed by utilizing SAB to capture semantic information among these substructures, thereby obtaining the feature representation of the drugs themselves. (2) Employing an MLP to acquire features of cancer cell lines based on normalized gene data. (3) Leveraging CAB to capture interaction features between drugs and cancer cell lines. Finally, we will aggregate these interaction features with the intrinsic features through a fusion coefficient.

3.2.1. Drug representation

For drug substructures obtained in data preprocessing, we construct a transformer encoder, which comprises embedding layers and self-attention blocks, to capture their interdependencies and obtain effective representations. Encoding layers encode the content and position of substructures separately, with the latter representing their position indices in the SMILES. Position encoding grants the transformer insight into the arrangement of drug substructures, thereby deepening

its understanding of drug structures. The self-attention blocks primarily focus on inter-substructure interactions to refine the embedding vectors, enhancing the model's comprehension of chemical semantic relationships.

Similar to constructing word vectors in NLP, we define substructure content and position index matrices, $M_c \in \mathbb{R}^{d \times k}$ and $M_p \in \mathbb{R}^{d \times d}$, along with content and position query dictionaries, $T_c \in \mathbb{R}^{k \times c}$ and $T_p \in \mathbb{R}^{d \times c}$, where d , k and c represent the maximum number of substructures for a single drug, the total number of substructures, and the dimensionality of the embedding vectors, respectively. We retrieve the corresponding embeddings from the dictionaries based on the indices, and then combine the content and position encodings to obtain the encoding of drug substructures as follows:

$$D_e = M_c T_c + M_p T_p. \quad (2)$$

Subsequently, D_e is inputted into the SAB with a multi-head attention machine. Each head has its query Q^i , key K^i , and value V^i , allowing SAB to focus on different aspects of drug substructures. Following this, SAB computes attention scores for each head, collects their respective outputs, and then concatenates them. The process is as follows:

$$Q^i = D_e W_q^i, K^i = D_e W_k^i, V^i = D_e W_v^i, \quad (3)$$

$$Att_i = \text{Softmax}\left(\frac{Q^i K^{iT}}{\sqrt{c/h}}\right) V^i, \quad (4)$$

$$Att = \text{Concat}(Att_1, \dots, Att_h) W_o, \quad (5)$$

where W_q^i , W_k^i , W_v^i , W_o are learnable weight matrices, and i , c , h represent the index for attention head, embedding dimensions, and the number of attention heads, respectively.

Finally, Att is fed into a fully connected feed-forward layer, obtaining drugs' intrinsic features D_f as follows:

$$D_f = \text{ReLU}(Att W_{f1} + \mathbf{b}_{f1}) W_{f2} + \mathbf{b}_{f2} \quad (6)$$

where W_{f1} and W_{f2} are weight parameters, and $\mathbf{b}_{f1}$ and $\mathbf{b}_{f2}$ are biases.

3.2.2. Cell line representation

The high-dimensional gene data of cell lines contains rich information. Previous studies (Jiang et al., 2022; Zhan et al., 2023) have employed MLP for dimensionality reduction to extract cell lines' features. In our work, gene expression and CNV features are inputted separately into the MLP with four fully connected layers. Subsequently, the features from these two parts are concatenated as cell line feature C_f :

$$C_1 = \text{MLP}(x_{\text{expr}}), C_2 = \text{MLP}(x_{\text{cnv}}), \quad (7)$$

$$\text{MLP}(x) = \sigma\left(\sigma\left(\sigma\left(W_1 x + \mathbf{b}_1\right) W_2 + \mathbf{b}_2\right) W_3 + \mathbf{b}_3\right) W_4 + \mathbf{b}_4, \quad (8)$$

$$C_f = \text{Concat}(C_1, C_2), \quad (9)$$

where x_{expr} and x_{cnv} represent the gene expression and CNV data of cell lines, respectively, and σ denotes a non-linear activation function. Additionally, W_1 , W_2 , W_3 , and W_4 are weight parameters, $\mathbf{b}_1$, $\mathbf{b}_2$, $\mathbf{b}_3$, and $\mathbf{b}_4$ are biases.

3.2.3. Interaction features and aggregation

At this point, we have obtained the intrinsic characteristics of drugs and cell lines. However, for predicting CDR, the interaction features are equally crucial. In this work, we constructed a CAB that uses highly asymmetric attention mechanisms. It takes the drug and cell line embeddings D_e and C_f as inputs to acquire their interaction information D_i and C_i , respectively. The specific calculation process is as follows:

$$\begin{aligned} Q^d &= D_e W_q^d, & K^d &= D_e W_k^d, & V^d &= D_e W_v^d, \\ Q^c &= C_f W_q^c, & K^c &= C_f W_k^c, & V^c &= C_f W_v^c, \end{aligned} \quad (10)$$

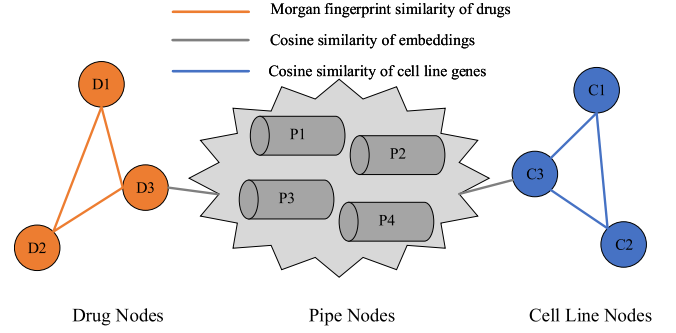


Fig. 2. A simple example of our association graph. The graph contains three types of nodes: drug nodes, pipe nodes, and cell line nodes. And edges are constructed between them based on similarity.

$$Att^d = \text{Softmax}\left(\frac{Q^d K^{dT}}{\sqrt{c}}\right) V^d, \quad (11)$$

$$Att^c = \text{Softmax}\left(\frac{Q^c K^{dT}}{\sqrt{c}}\right) V^c, \quad (12)$$

$$\begin{aligned} D_i &= \text{MLP}(Att^d), \\ C_i &= \text{MLP}(Att^c), \end{aligned} \quad (13)$$

where W_q^d , W_k^d , W_v^d , W_q^c , W_k^c and W_v^c are learnable weight parameters.

Following that, we introduce a fusion coefficient β to aggregate the intrinsic features with the interaction features, thereby obtaining the fine-grained representation D , C of drugs and cell lines as follows:

$$\begin{aligned} D &= (1 - \beta) D_f + \beta D_i, \\ C &= (1 - \beta) C_f + \beta C_i. \end{aligned} \quad (13)$$

3.3. Association graph module

After obtaining the fine-grained features of drugs and cell lines, the subsequent step is acquiring their coarse-grained features at the network level. To achieve this, we construct an association graph based on the similarity between nodes (Fig. 2). Such an approach has demonstrated effectiveness in predicting associations between heterogeneous substances (Wu et al., 2022). Specifically, within the heterogeneous domain, we introduce pipe nodes as mediators linking drug nodes and cell line nodes to further bridge the gap of features between drugs and cell lines. The pipe node, denoted as a c -dimensional vector P^k , is initially distributed according to a normal distribution. We utilize cosine similarity to define the edges between pipe nodes and drug or cell line nodes:

$$\begin{aligned} E_{d_i, p_k} &= \frac{D^i \cdot P^k}{\|D^i\| \cdot \|P^k\|}, \\ E_{c_j, p_k} &= \frac{C^j \cdot P^k}{\|C^j\| \cdot \|P^k\|}, \end{aligned} \quad (14)$$

where D^i and C^j represent the fine-grained features of drug i and cell line j , respectively. The symbol $\cdot$ denotes the inner product of two vectors, while $\|\cdot\|_2$ represents the L_2 norm of a vector.

In the homogeneous region, if we follow the above method to establish edges, the accuracy of similarity calculation depends on the quality of our obtained features from Sec 2.2.3. To address this issue, we utilize the drug-drug similarity and the cell line-cell line similarity as defined in Section 2.1, which are both calculated based on the original substances:

$$E_{d_i, d_n} = \frac{a_{in}}{a_i + a_n - a_{in}}, \quad (15)$$

$$E_{c_j, c_m} = \frac{1}{N} \sum_{n=1}^N \frac{x_j^n \cdot x_m^n}{\|x_j^n\| \cdot \|x_m^n\|}, \quad (16)$$

where a_i , a_n , and a_{in} represent the number of non-zero bits in the Morgan fingerprints of drug i , drug n , and their shared non-zero bits, respectively. N is the number of genes in the cell line, while x_j^n and x_m^n represent the normalized feature vectors of the n th gene in cell lines j and m , respectively.

We construct the adjacency matrix $A_{ij} \in \mathbb{R}^{(2+m+2n) \times (2+m+2n)}$ for drug i and cell line j by selecting m pipe nodes, n drug nodes, and n cell line nodes as follows:

$$A_{ij} = \begin{pmatrix} & & & DD & DC_1 \\ & A'_{ij} & & DC_2 & CC \\ \hline DD^T & DC_1^T & & PDC & \\ DC_2^T & CC^T & & & DC_3 \end{pmatrix}, \quad (17)$$

where

$$A'_{ij} = \begin{pmatrix} E_{d_i,d_i} & E_{d_i,c_j} & E_{d_i,p_1} & \dots & E_{d_i,p_m} \\ E_{c_j,d_i} & E_{c_j,c_j} & E_{c_j,p_1} & \dots & E_{c_j,p_m} \\ E_{p_1,d_i} & E_{p_1,c_j} & E_{p_1,p_1} & \dots & E_{p_1,p_m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ E_{p_m,d_i} & E_{p_m,c_j} & E_{p_m,p_1} & \dots & E_{p_m,p_m} \end{pmatrix}, \quad (18)$$

$$DD = (E_{d_i,d_1} \quad E_{d_i,d_2} \quad \dots \quad E_{d_i,d_n}), \quad (19)$$

$$CC = (E_{c_j,c_1} \quad E_{c_j,c_2} \quad \dots \quad E_{c_j,c_n}), \quad (20)$$

$$DC_1 = (E_{d_i,c_1} \quad E_{d_i,c_2} \quad \dots \quad E_{d_i,c_n}), \quad (21)$$

$$DC_2 = (E_{c_j,d_1} \quad E_{c_j,d_2} \quad \dots \quad E_{c_j,d_n}), \quad (22)$$

$$PDC = \begin{pmatrix} E_{p_1,d_1} & \dots & E_{p_1,d_n} & E_{p_1,c_1} & \dots & E_{p_1,c_n} \\ E_{p_2,d_1} & \dots & E_{p_2,d_n} & E_{p_2,c_1} & \dots & E_{p_2,c_n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ E_{p_m,d_1} & \dots & E_{p_m,d_n} & E_{p_m,c_1} & \dots & E_{p_m,c_n} \end{pmatrix}, \quad (23)$$

$$DC_3 = \begin{pmatrix} E_{d_1,d_1} & \dots & E_{d_1,d_n} & E_{d_1,c_1} & \dots & E_{d_1,c_n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ E_{c_1,d_1} & \dots & E_{c_1,d_n} & E_{c_1,c_1} & \dots & E_{c_1,c_n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ E_{c_n,d_1} & \dots & E_{c_n,d_n} & E_{c_n,c_1} & \dots & E_{c_n,c_n} \end{pmatrix}. \quad (24)$$

To better explore the association between drug i and cell line j , and avoid the over-smoothing issue in GNN, we set DC_1 , DC_2 , DC_3 , and PDC as zero matrices. This implies that the homogeneous neighbor nodes of drug i and cell line j solely serve to enhance similarity. Then, we utilize a three-layer GNN to capture network-level features on the above association graph:

$$X_{ij}^0 = (D^i, C^j, P^1, \dots, D^1, \dots, C^1, \dots, C^n)^T, \quad (25)$$

$$X_{ij}^{l+1} = \text{ReLU}(\text{ReLU}(D_{ij}^{-\frac{1}{2}} A_{ij} D_{ij}^{-\frac{1}{2}}) X_{ij}^l W_{ij}^l + b_{ij}^l), \quad (26)$$

where X_{ij}^0 represents the initial features of nodes in the association graph, and X_{ij}^l denotes the node features after l layers of information propagation. D_{ij} is the degree matrix of the adjacency matrix A_{ij} . W_{ij}^l and b_{ij}^l represent the weight and bias parameters of the l th layer of the GNN.

Finally, we extract the first two rows of X_{ij}^3 as the network-level features $\hat{D}^i$ and $\hat{C}^j$ for drug i and cell line j . We concatenate the

fine-grained features with the network-level features to form the final drug-cell line features $\overline{D}^i$ and $\overline{C}^j$:

$$\begin{aligned} \overline{D}^i &= \text{Concat}(D^i, \hat{D}^i), \\ \overline{C}^j &= \text{Concat}(C^j, \hat{C}^j). \end{aligned} \quad (27)$$

3.4. Predict module

After obtaining the combined features of fine-grained and network-level features, the final step is to predict the $\ln(\text{IC50})$ values. We first simulate the interaction between the drug and cell line by element-wise multiplication:

$$I_{ij} = \overline{D}^i \odot \overline{C}^j. \quad (28)$$

Subsequently, we feed I_{ij} into a two-layer perceptron to obtain the predicted $\ln(\text{IC50})$ values $\hat{y}_{ij}$ as follows:

$$\hat{y}_{ij} = (\text{ReLU}(I_{ij} W_p^1 + b_p^1)) W_p^2 + b_p^2 \quad (29)$$

where W_p^1 and W_p^2 represent weight parameters, while b_p^1 and b_p^2 are bias parameters.

3.5. Loss function

In our experiment, we leverage the Mean Squared Error (MSE) as the loss function to quantify the squared differences between the predicted $\ln(\text{IC50})$ values and the ground truth:

$$\text{Loss} = \frac{1}{|\mathcal{N}|} \sum_{(i,j) \in \mathcal{N}} (\hat{y}_{ij} - y_{ij})^2, \quad (30)$$

where $\mathcal{N}$ represents the set of drug-cell line pairs in the training dataset, and y_{ij} denotes the true $\ln(\text{IC50})$ value of drug i with cell line j .

4. Results and discussion

4.1. Implementation details

FA²GL is implemented in the PyTorch framework. For the experiment, we employ the Adam optimizer (Kingma and Ba, 2014) with a learning rate of 0.0001 and set the penalty coefficient λ for the L2 regularization term to 0.01. Training consists of 200 epochs, during which drug-cell line pairs are randomly shuffled, and the batch size is set to 32. In addition, other hyperparameters used during the training process are shown in Table 1.

4.2. Datasets

In this study, we utilize the dataset provided by DualGCN (Ma et al., 2022). This dataset is derived from the GDSC database, from which we extract PubChem IDs, cell line names, and corresponding IC50 values. Additionally, we obtain the gene data of the cell lines, including gene expression and CNV, from the CCLE database.

After obtaining the raw data from GDSC, a screening process is conducted. For drugs, only those with exactly one PubChem and GDSC identifier are retained. For cell lines, only those in the CCLE dataset with complete gene expression, CNV data, and cancer-type annotations are preserved. Additionally, cell lines with fewer than 10 samples per cancer type are filtered out. Ultimately, we retain a total of 86,530 instances, covering 208 drugs and 525 cell lines across 27 cancer types.

Table 1
The hyper-parameters in FA²GL.

Hyper-parameter		Value
Drug	Num of substructures	50
	Num of self-transformer layers	3
	Num of attention heads	8
Cell line	Neuron num of MLP _c	512, 256, 128, 100
Aggregation	Fusion coefficient β	0.1
Association graph	Num of pipe nodes	64
	Num of drugs nodes	32
	Num of cell line nodes	32
	Num of GNN layers	3
	Neuron num of GNN	128, 128, 128
Prediction	Neuron num of MLP _p	128, 1

Table 2
Performance comparison results.

Method	Pearson's correlation	Spearman's correlation	RMSE
Random Forest	0.864 ± 0.001	0.839 ± 0.003	1.441 ± 0.008
Lasso	0.893 ± 0.002	0.873 ± 0.002	1.284 ± 0.007
DeepCDR(-)	0.900 ± 0.004	0.877 ± 0.004	1.265 ± 0.020
CDRscan	0.911 ± 0.002	0.894 ± 0.002	1.173 ± 0.011
DualGCN	0.925 ± 0.001	0.907 ± 0.002	1.079 ± 0.007
DeepCDR	0.928 ± 0.001	0.910 ± 0.001	1.066 ± 0.004
BANDRP	0.927 ± 0.003	0.909 ± 0.002	1.016 ± 0.013
FA ² GL	0.934 ± 0.002	0.914 ± 0.001	1.031 ± 0.003

4.3. Evaluation framework and metrics

We conduct five-fold cross-validation on the dataset. The experimental result is obtained by averaging the results across the five folds. Furthermore, to evaluate model performance, we utilize three widely adopted regression task evaluation metrics: Pearson's correlation coefficient (PCC), Spearman's rank correlation coefficient (SCC), and Root Mean Squared Error (RMSE). The calculation formulas for these metrics are as follows:

$$\text{PCC} = \frac{\sum_{(i,j) \in \mathcal{N}} (\hat{y}_{ij} - \bar{\hat{y}})(y_{ij} - \bar{y})}{\sqrt{\sum_{(i,j) \in \mathcal{N}} (\hat{y}_{ij} - \bar{\hat{y}})^2} \sqrt{\sum_{(i,j) \in \mathcal{N}} (y_{ij} - \bar{y})^2}} \quad (31)$$

$$\text{SCC} = 1 - \frac{6 \sum_{(i,j) \in \mathcal{N}} d_{ij}^2}{|\mathcal{N}|(|\mathcal{N}|^2 - 1)} \quad (32)$$

$$\text{RMSE} = \sqrt{\frac{1}{|\mathcal{N}|} \sum_{(i,j) \in \mathcal{N}} (\hat{y}_{ij} - y_{ij})^2} \quad (33)$$

where $\bar{\hat{y}}$ and $\bar{y}$ are the means of all predicted values $\hat{y}_{ij}$ and true values y_{ij} , respectively. And d_{ij} represents the difference in ranks between $\hat{y}_{ij}$ and y_{ij} .

4.4. Performance comparison of our method and existing methods

To evaluate the performance of our model, we compare it with the following baseline models:

- **CDRscan** (Chang et al., 2018) utilized a convolutional neural network to extract drug molecular fingerprints and cell line genomic mutation. We adopted the optimal parameters provided in the original paper.
- **DeepCPR** (Liu et al., 2020) integrated multi-omics features of genes, including SNV, insertion/deletion mutations, gene expression, and DNA methylation. It employed a hybrid graph convolutional network to separately obtain graph features of drugs and cell lines. We refer to the DeepCPR model without SNV features as DeepCPR(-). For hyperparameter tuning, we explore the number

of GCN layers h with values [1, 3, 5] and the embedding size q with values [128, 256, 512]. We select $h = 3$, $q = 256$ as the optimal parameters.

- **DualGCN** (Ma et al., 2022) constructed a gene interaction network based on protein-protein interactions. It then utilized a unified dual graph convolutional network to encode drugs and cell lines on drug molecule graphs and above gene interaction network, avoiding using SNV data. Since we use the same dataset as the GCN paper, we adopt the default hyperparameters specified in the paper: Drug GCN with two layers, with dimensions 256 and 128, and Cell Line GCN with four layers, with a dimension of 256. **BANDRP** (Cao et al., 2024) is a bilinear attention model that integrates multi-omics data of cancer cell lines and multiple molecular fingerprints of drugs to predict anti-cancer drug response with enhanced feature representation and interaction learning. We unify the feature representation data sources for both cell lines and drugs and tune the embedding size of drugs and cell lines h_d and h_c with values [56, 128, 256, 512] as well as the number of bilinear attention heads n_c from [1,3,5]. We select $h_d = 128$, $h_c = 128$ and $n_c = 3$ as the optimal parameters.
- **Random Forest** is an ensemble learning method based on multiple decision trees. We concatenate drug atomic features and cell line genetic features as input. During training, we select the number of decision trees n_t , from the set [50, 100, 200], with corresponding RMSE values of 1.445, 1.441, and 1.441, respectively. Hence, we set n_t to 100.
- **Lasso**, a variant of linear regression, added an L1 regularization term to the loss function. The input is consistent with the aforementioned method. During training, we experimented with different hyperparameter settings, such as the regularization parameter α , selected from [0.01, 0.1, 0.5], with RMSE values of 1.284, 1.715, and 2.556, respectively. Therefore, we set α to 0.01.

Table 2 shows that FA²GL achieves outstanding predictive performance, with a Pearson correlation coefficient of 0.934, a Spearman correlation coefficient of 0.914, and an RMSE of 1.031. Compared to traditional methods such as random forest and Lasso, FA²GL demonstrates significant improvements. Furthermore, among deep learning-based methods, FA²GL not only outperforms BANDRP, DeepCPR(-), CDRscan, and DualGCN (without SNV data) in correlation-based metrics but also surpasses DeepCPR, which utilizes extensive SNV data, across all evaluation criteria. In terms of RMSE, a deterministic metric, our model ranks second, slightly behind BANDRP. In the task of predicting anti-cancer drug response, correlation metrics reflect the ranking relationship between IC50 values and drug efficacy against specific cancer cell lines. From a practical perspective, identifying the most effective drugs for a given cancer cell line is often more crucial than predicting absolute IC50 values, though the latter remains important. While BANDRP performs well in absolute IC50 prediction, it does not excel in correlation-based metrics, likely due to its lack of network-level feature extraction. In contrast, our approach leverages a broader range of drug-cell line interactions, enabling a more comprehensive learning of holistic features, thereby achieving superior performance in ranking-based evaluations.

Additionally, we adopt a cautious approach to SNV data usage for two primary reasons. First, different research projects employ diverse methods for SNV data acquisition and processing, and they use different reference genomes, leading to potential challenges in aligning SNV data from various sources (Ma et al., 2022). Second, although our study utilizes in vitro cancer cell lines, single-cell and clinical data provide a more accurate reflection of tumor biology. However, these data types generally have lower sequencing coverage and depth, making SNV calls less reliable. Despite not relying heavily on extensive SNV data, FA²GL still achieves high predictive accuracy and has strong potential for extension to single-cell and clinical datasets, contributing further to advancements in precision medicine.

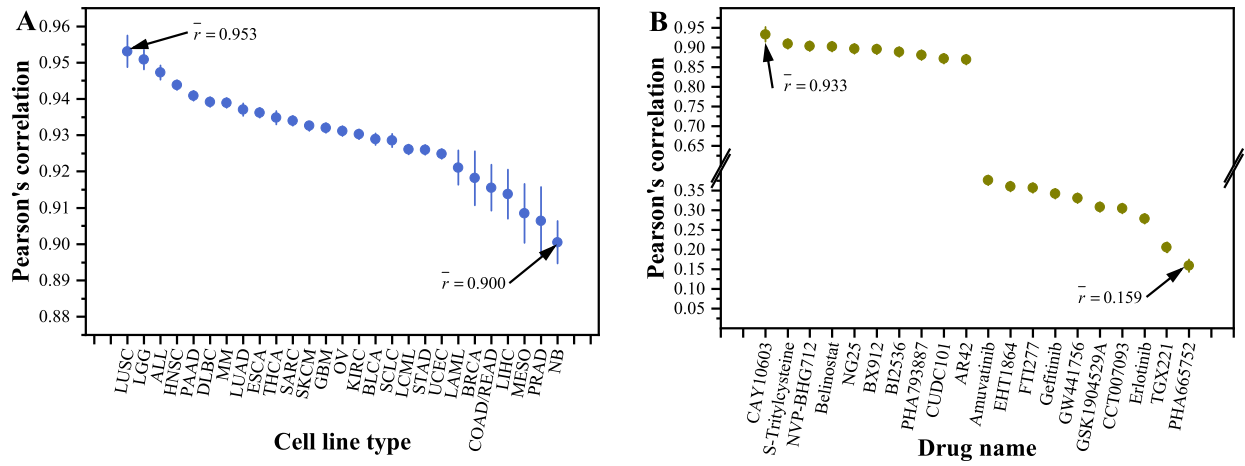


Fig. 3. Performance of FA²GL on different cell line types and drugs. A. Average Pearson correlation coefficients for each type of cell line, arranged from left to right in descending order. Blue dots represent the average Pearson correlation coefficients under different cross-validation (CV) settings, while the blue vertical lines indicate the error bars. LUSC and NB exhibit the best and worst performance, with average Pearson coefficients of 0.953 and 0.900, respectively. B. Average Pearson correlation coefficients for the top and bottom 10 drugs, arranged from largest to smallest. The meaning of green dots and vertical lines is the same as described above. CAY10603 and PHA665752 exhibit the best and worst performance, with average Pearson coefficients of 0.933 and 0.159, respectively.

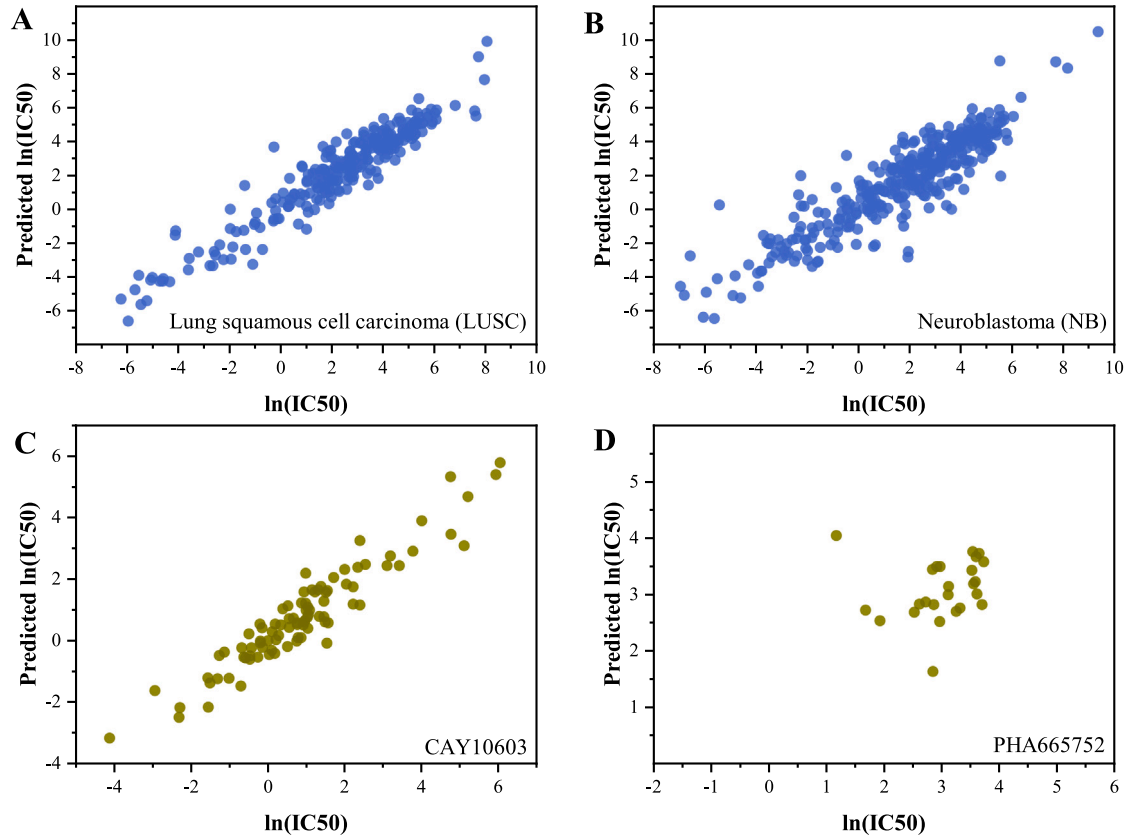


Fig. 4. Scatter plot of true $\ln(\text{IC}_{50})$ versus predicted $\ln(\text{IC}_{50})$ for cell lines and drugs with the highest and lowest average Pearson correlation coefficients. A. LUSC: Cell line with the highest average Pearson correlation coefficient. B. NB: Cell line with the lowest average Pearson correlation coefficient. C. CAY10603: Drug with the highest average Pearson correlation coefficient. D. PHA66575: Drug with the lowest average Pearson correlation coefficient.

4.5. Predicting cell lines and drugs

To assess FA²GL's predictive performance across diverse cancer cell lines, we partitioned the dataset by cancer type, assigning 80% of samples from each cancer type to the training set and the remaining 20% to the testing set. Employing five-fold cross-validation, we

averaged the results over five runs. Fig. 3A illustrates the Pearson correlation coefficients across 27 cell lines, ranging from 0.953 to 0.900. The highest and lowest coefficients are obtained by the LUSC and NB cell lines, respectively. Corresponding scatter plots for these cell lines are provided in Fig. 4A, B. Above results highlight FA²GL's robust predictive ability across diverse cell lines. Furthermore, we evaluated

Table 3
Results of attention scores and ranking.

	CID	Drug name	Substructure	Cell line name	Attention score	Ranking
Resistant	11 617 559	GW2580	"2=C(C"	ACH-000392	0.922	2/17
	92 577	NSC87877	"2=C(C"	ACH-000392	0.609	4/19
	9 813 758	BMS345541	"2=C(C"	ACH-000392	0.553	3/11
	53 394 750	–	"2=C(C"	ACH-000392	0.683	3/17
Sensitive	24 826 799	PONATINIB	"4=C"	ACH-000198	0.632	3/20
	53 302 361	HG6-64-1	"4=C"	ACH-000198	0.968	2/24
	49 836 027	Torin1	"4=C"	ACH-000198	0.563	3/27

the predictive performance of various drugs, with Fig. 3B displaying the Pearson correlation coefficients for the top and bottom 10 drugs. The performance varied significantly, with correlation coefficients ranging from 0.933 to 0.159, due to differences in the amount of data used to train the models for different drugs. Specifically, CAY10603 attained 0.933 in the top-performing drugs, contrasting with PHA66575's 0.159 in the bottom-performing ones. Scatter plots for these drugs are depicted in Fig. 4C, D.

4.6. Visualization of the relationship between drug substructures and cancer cell lines

To analyze the relationship between drug substructures and cell lines, we visualize their attention scores in CAB. Specifically, we focus on the drug NG25 and randomly selected 10 samples from cell lines within each $\ln(\text{IC}_{50})$ interval: $(-6, -3]$, $(-3, 0]$, $(0, 2]$, $(2, 3]$, $(3, 5]$, and $(5, 6]$. Fig. 5 illustrates their attention scores where cell lines are arranged from left to right in ascending order of $\ln(\text{IC}_{50})$. The gray box denotes cell lines with $\ln(\text{IC}_{50})$ values in the $(-6, -3]$ range, categorized as the sensitive region, while the orange box signifies those in the $(5, 6]$ range, categorized as the resistant region. We observe that regardless of the region, cell lines show higher attention scores for at least one substructure of drug NG25. We then select a pair of substructure-cell line with high attention scores from each region to test if this pattern holds for other drugs. Specifically, we choose NG25's third substructure "2=C(C" and cell line 59 (ACH-000392) in the resistant region, and the fifteenth substructure "4=C" and cell line 2 (ACH-000198) in the sensitive region, all marked with a yellow triangle in Fig. 5. Next, we screen other drugs meeting the following criteria, respectively: having an $\ln(\text{IC}_{50}) > 5$ with ACH-000392 and containing the specific substructure "2=C(C", as well as those with an $\ln(\text{IC}_{50}) < -4$ in relation to ACH-000198 and containing the substructure "4=C". We have identified 4 drugs meeting the first criterion and 3 drugs meeting the second criterion. Finally, We not only obtain the attention scores between above drugs' specified substructure and cell lines, but also determine the ranking of the attention scores of the aforementioned substructures among all substructures. The result shown in Table 3 demonstrates that the patterns observed in NG50 are consistent across other drugs, suggesting that the FA²GL effectively captures the intrinsic relationships between drug substructures and cell lines.

4.7. Ablation experiments

We conduct ablation experiments with the following three settings:

- FA²GL/SAB: Exclude the self-attention block that refines the embedding of drug substructures.
- FA²GL/CAB: Remove the cross-attention block which captures interaction information between drugs and cell lines.
- FA²GL/AGM: Eliminate the association graph module and directly feed the embeddings of drugs and cell lines obtained from the previous module into the predict module.

Table 4 indicates that the performance of FA²GL declines to varying degrees when the mentioned block or module is absent. We observe a more significant decrease in performance when AGM is missing

Table 4
Ablation experiment results.

Method	PCC	SCC	RMSE
FA ² GL/SAB	0.930 ± 0.001	0.908 ± 0.004	1.060 ± 0.002
FA ² GL/CAB	0.928 ± 0.003	0.910 ± 0.002	1.052 ± 0.003
FA ² GL/AGM	0.926 ± 0.002	0.908 ± 0.004	1.093 ± 0.001
FA ² GL	0.934 ± 0.002	0.914 ± 0.001	1.031 ± 0.003

compared to the absence of CAB. We speculate that despite the absence of CAB, the pipeline nodes introduced in AGM can still transmit and integrate features of drugs and cell lines. However, once AGM is removed, FA²GL will be unable to utilize similarity information between substances, leading to a decrease in performance. As for SAB, we believe it influences drug representation by learning relationships between drug substructures, thereby affecting the model's performance.

5. Conclusion

Anti-cancer drugs play a crucial role in cancer treatment. Predicting the response between drugs and cancer cell lines is of great importance for accelerating the development of anti-cancer drugs. Currently, many computational methods have been based on in vitro cancer cell lines to study CDR.

In this paper, we propose a novel model called FA²GL based on feature aggregation and association network learning. This model not only obtains fine-refined features of drugs and cell lines through SAB, MLP and CAB blocks, but also considers network-level features by constructing an association graph. Moreover, FA²GL avoids using large-scale gene SNV features, thus having the potential to be extended to clinical patients or single-cell levels, promoting the development of precision medicine. Through experimental validation, FA²GL without SNV features demonstrates excellent its predictive performance.

However, our model still has some limitations. For instance, the source of cell line features is relatively limited, considering only partial gene information, while ignoring some non-genetic features such as cell pathways and histone modifications. In future research, we will delve deeper into integrating multi-omics information to better characterize cell line features. Moreover, while FA²GL is capable of capturing the correlation between drug substructures and cell lines, further exploration of the underlying biological principles are waiting to be undertaken.

CRedit authorship contribution statement

Kaiyi Xu: Writing – original draft, Visualization, Validation, Conceptualization. **Minhui Wang:** Data curation. **Xin Zou:** Supervision. **Chengfu Ji:** Data curation. **Hua Zhou:** Data curation. **Chang Tang:** Supervision, Methodology, Funding acquisition.

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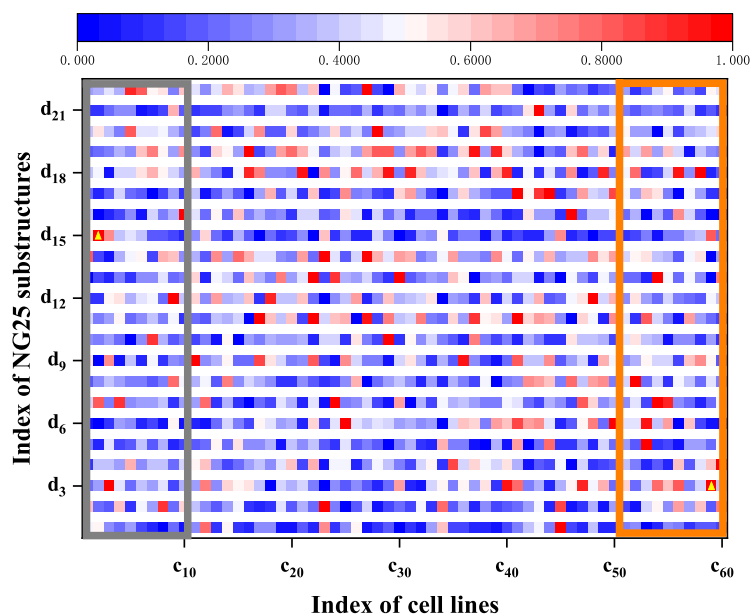


Fig. 5. The heatmap of attention scores between the substructures of drug NG50 and various cell lines, where the gray box represents the sensitive region ($\ln(\text{IC}_{50})$ ranging from -6 to -3), while the orange box signifies the resistant region ($\ln(\text{IC}_{50})$ ranging from 5 to 6). Additionally, two yellow triangles mark the attention scores between the third substructure “2=C(C)” of NG50 and cell line 59 (ACH-000392) in the resistant region and the fifteenth substructure “4=C” of NG50 and cell line 2 (ACH-000198) in the sensitive region. These serves as case studies for our in-depth exploration of the relationship between drug substructures and cell lines.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Datas and codes utilized in this work are accessible for free at: <https://github.com/Dtdtxuky/FA-2GL/tree/main>.

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